—something that physicists call a *background-independent* formulation (the term comes from the loose notion of spacetime as a backdrop against which physical phenomena take place). Instead, essentially all approaches envision strings as moving and vibrating through a spacetime that is inserted into the theory "by hand"; spacetime does not emerge from the theory, as physicists imagine it would in a background-independent framework, but is supplied to the theory by the theorist. Many researchers consider the development of a background-independent formulation to be the single greatest unsolved problem facing string theory. Not only would it give insight into the origin of spacetime, but a background-independent framework would likely be instrumental in resolving the major hang-up encountered at the end of Chapter 12—the theory's current inability to select the geometrical form of the extra dimensions. Once its basic mathematical formalism is disentangled from any particular spacetime, the reasoning goes, string theory should have the capacity to survey all possibilities and perhaps adjudicate among them.

Another difficulty facing the strings-as-threads-of-spacetime proposal is that, as we learned in Chapter 13, string theory has other ingredients besides strings. What role do these other components play in spacetime's fundamental makeup? This question is brought into especially sharp relief by the braneworld scenario. If the three-dimensional space we experience is a three-brane, is the brane itself indecomposable or is it made from combining the theory's other ingredients? Are branes, for example, made from strings, or are branes and strings both elementary? Or should we consider yet another possibility, that branes and strings might be made from some yet finer ingredients? These questions are at the forefront of current research, but since this final chapter is about hints and clues, let me note one relevant insight that has garnered much attention.

Earlier, we talked about the various branes one finds in string/M-THEORY: one-branes, two-branes, three-branes, four-branes, and so on. Although I didn't stress it earlier, the theory also contains *zero-branes*—ingredients that have no spatial extent, much like point particles. This might seem counter to the whole spirit of string/M-theory, which moved away from the point-particle framework in an effort to tame the wild undulations of quantum gravity. However, the zero-branes, just like their